



Om Sharma

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B.Tech - Mechanical Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology, Guwahati	8.30 (Current)	2025-Present
Senior Secondary (XII)	CBSE Board	86.2%	2025
Secondary (X)	CBSE Board	93.6%	2023

PROJECTS

- GradeOps: AI-Powered Human-in-the-Loop Exam Grading Platform** *Mar. 2025 - Present*
Coding Club, IIT Guwahati · Python, FastAPI, React, LangGraph, Gemini, Nemotron, PostgreSQL *Github*
 - Built a platform that **OCRs handwritten exam scripts via NVIDIA Nemotron**, scores each answer against instructor rubrics using **Google Gemini**, and routes every result through a **human review queue** before grades are released.
 - Engineered a **LangGraph state machine** orchestrating the two-stage AI pipeline with **SQLite checkpointing** so pending reviews survive server restarts; backend served via **FastAPI on PostgreSQL** with role-based cookie auth.
 - Developed the **React frontend from scratch** – Instructor and TA dashboards, keyboard-friendly review queue, and a premium animated landing page with **Three.js** visuals.
- Delhivery Network Intelligence: Graph-Based Logistics Optimization** *Jun. 2025*
Consulting & Analytics Club, IIT Guwahati · Python, XGBoost, NetworkX, node2vec, Pandas, Streamlit *Github*
 - Modelled Delhivery's logistics network as a directed weighted graph on 144,900 scans to identify that 5 hubs account for 30.8% of total network excess delay; top bottleneck carries 12.5% of network delay alone.
 - Built graph-enhanced XGBoost ETA model using node2vec hub embeddings and betweenness centrality features, achieving **MAE 28 min vs OSRM's 108 min (74% improvement)** and 58.3% on-target accuracy vs 4.1%.
 - Delivered FTL vs Carting decision framework and upgrade impact analysis estimating **28% reduction in late deliveries** and **~Rs. 1.5 cr/yr** recovered revenue from top-3 hub upgrades.
- Autonomous Bidding Agent for Used-Car Auctions** *Apr. 2025*
Coding Club Recruitment, IIT Guwahati · Python, XGBoost, Pandas, Scikit-learn *Github*
 - Trained XGBoost regressor (log-target transform, engineered features: age, usage intensity, condition ratio) on historical US auction data; tuned via RandomizedSearchCV achieving **Val RMSE \$1,620 – 14.3% improvement** over baseline.
 - Designed sigmoid-proportional bidding strategy with bankroll-adaptive ceiling and hard cap at 95% of predicted value, structurally guaranteeing profit on every win; deployed as a live auction simulation agent.
- Bangladesh Power Demand Forecasting** *Apr. 2025*
.ai Club, IIT Guwahati · Python, Scikit-learn, Random Forest, Pandas, NumPy *Github*
 - Fused PGCB power records, Dhaka weather observations, and World Bank economic indicators (~92,000 rows, 10-year span) into an end-to-end ML pipeline for hourly electricity demand forecasting.
 - Engineered autoregressive lag features (t-1h, t-24h, t-168h) and cyclical time encodings; tuned Random Forest via GridSearchCV with TimeSeriesSplit CV, achieving **MAPE < 3%** and **R² > 0.97** on chronological hold-out.

TECHNICAL SKILLS

- Programming Languages:** C/C++, Python
- Libraries & Tools:** XGBoost, Scikit-learn, Pandas, NumPy, NetworkX, node2vec, Streamlit, Git, L^AT_EX
- Key Concepts:** Machine Learning, Graph Analytics, Time Series Forecasting

POSITIONS OF RESPONSIBILITY

- Deputy Coordinator**, Coding Club, IIT Guwahati (Machine Learning Module) *Jun. 2026 - Present*
- Associate**, Finance & Economics Club, IIT Guwahati *Jun. 2026 - Present*
- Senior Executive**, Entrepreneurship Cell (E-Cell), IIT Guwahati *Oct. 2025 - Present*
- Associate**, Aeromodelling Club, IIT Guwahati *Feb. 2026 - Present*
- Associate**, Robotics Club, IIT Guwahati *Feb. 2026 - Present*

KEY COURSES TAKEN

- Programming and Data Structures, Linear Algebra, Multivariable Calculus, Real Analysis, Complex Analysis, Machine Learning Specialization (Andrew Ng)

ACHIEVEMENTS

- Codeforces**, Rating 693, 160+ problems solved — Greedy, Binary Search & DP *2025-Present*